International Trade[1]

Lecture Note 3: Ricardo with a Continuum of Goods and n Countries

Martín Alfaro
University of Alberta

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[1]The notes are still preliminary and in beta. Please, if you find any typo or mistake, send it to malfaro@ualberta.ca.

Contents

Notation

This is a derivation

This is some comment

This is a comment on advanced topics which are not part of the course (you can ignore it without loss of continuity regarding the text)

- The symbol ":=" means "by definition".
- Vectors are denoted by bold lowercase letters (for instance, $\mathbf{x}$) and matrices by bold capital letters (for instance, $\mathbf{X}$).
- To differentiate between the verb "maximize" and the operator "maximum", I denote the former with "max" and the latter with "sup" (i.e., supremum). The same notation applies to "minimize" and "minimum", where I use "min" and "inf" (the latter indicates infimum).
- "iff" means "if and only if"
- $\exp(x)$ is the function e^x.
- Random variables are denoted with a bar below. For instance, $\underline{x}$.

These notes contain hyperlinks in blue and red. If you are using Adobe Acrobat Reader, you can click on the link and easily navigate back by pressing Alt+Left Arrow.

1 Problems of Generalization

Ricardo's original model is based on a simple framework involving two countries and two goods. In our previous lecture note, we expanded this model to accommodate a continuum of goods, while still assuming only two countries. This extended model yields similar insights into production patterns, where countries specialize and export goods according to their comparative advantages. Moreover, it can be proven that the same core insights remain valid in a model with multiple countries, provided there are only two goods. Overall, we can conclude that starting from a model with two countries and two goods, the assumption on the number of countries *or* goods can be easily relaxed, while still obtaining similar conclusions. The reason why this generalization is possible is that both variants allow for a clear ranking of each country's efficiency. Consequently, the mechanisms underlying these models are ultimately the same.

On the contrary, generalizing the Ricardian model to accommodate multiple goods and countries simultaneously is not straightforward. In fact, this generalization invalidates several of the conclusions obtained in Ricardo's original framework. For instance, (bilateral) comparative advantages alone are no longer sufficient to identify production and trade patterns.

To illustrate this issue, consider the following example from Jones (1961). He envisions a world with three countries and three goods, with the following unit labor requirements:

Figure 1: *Unit Labor Requirements*

Country	(A)	(B)	(C)
Good 1	10	10	10
Good 2	5	7	3
Good 3	4	3	2

Suppose the following pattern of specialization:

- (C) in good 3,
- (A) in good 2, and

- (B) in good 1.

Given these specializations, each country has a bilateral comparative advantage in the good it produces.[1] Nonetheless, it can be shown that this pattern of production cannot be part of a competitive solution under free trade.[2]

1.1 Jones (1961) (OPTIONAL)

Jones (1961) provides an approach to identifying the equilibrium when there is complete specialization in each country. He defines an ***assignment*** as a pattern of complete specialization, where labor in each country is completely allocated to the production of one good. He additionally defines a ***class of assignments***, as the collection of assignments with the same number of countries specialized in each good. Given these definitions, Jones (1961) identifies the optimal assignment in its class. Then, he pins down the equilibrium by using that a competitive solution needs to be optimal, as shown by McKenzie (1954).

The conclusions he obtains are the following. First, excluding cases where two countries have exactly the same relative costs, there is a unique optimal assignment in each class. Optimality is defined relative to the world efficiency locus. Additionally, the optimal solution needs to minimize the product of labor coefficients involved in the assignment. In a two-by-two world, this coincides with comparative advantages. More generally, though, while bilateral comparative advantages need to hold in the optimal allocation, this is not sufficient to get an equilibrium.

To see this, consider again the example in Figure 1. The pattern of specialization with bilateral comparative advantages considered above is indeed inefficient, even though it is optimal. And, by using the approach of Jones (1961), it can be shown that the optimal assignment of the class is (A) to good 1, (B) to good 3, and (C) to good 2. In that case,

[1] Given a pattern of production which is a candidate to an equilibrium, bilateral comparative advantages are defined according to the good that each country is producing. In other terms, bilateral comparative advantages do not require a comparison of relative efficiencies for goods different to the ones assigned to each country. In the example considered, let $a_i(j)$ be the unit labor requirement of country i for good j. Then, (C) relative to (A) satisfies $\frac{a_C(3)}{a_A(3)} = \frac{2}{4} < \frac{a_C(2)}{a_A(2)} = \frac{3}{5}$ and relative to (B), $\frac{a_C(3)}{a_B(3)} = \frac{2}{3} < \frac{a_C(1)}{a_A(1)} = \frac{10}{10}$, while (B) relative to (C), $\frac{a_B(1)}{a_C(1)} = \frac{10}{10} < \frac{a_B(3)}{a_C(3)} = \frac{3}{2}$. Thus each country has a bilateral comparative advantage in the good that it is producing.

[2] In particular, McKenzie (1954) shows that this pattern of production is not world efficient and, so, it can be shown that it cannot consittute a competitive solution.

the product of unit labor requirements is $10 \times 3 \times 3 = 90$, while in the other assignment would be $10 \times 5 \times 2 = 100$.

The set of optimal assignments determines the world efficiency frontier. And, given some demand conditions, one of the points along this frontier constitutes the actual production in a competitive solution.

2 Intuitions Behind EK

Eaton and Kortum (2002) (henceforth, EK) has had a profound impact on the trade literature over the past few decades. Their approach not only extends the DFS model to accommodate an arbitrary number of countries. but also provides a methodology to bring the Ricardian model to the data.

To understand their importance, a significant challenge in extending the Ricardian model to a multi-country setting is the loss of a natural ordering of goods. EK addresses this issue by employing a probabilistic approach to comparative advantages, which effectively renders the relative-efficiency ordering irrelevant for deriving results.

Preserving the relevance of comparative advantages is not without any cost: the model now cannot longer predict which *particular goods* each country will produce and trade. However, it's still possible to identify the *share* of goods produced by each country, as long as the *total* bilateral trade flows between countries. Additionally, he model remains capable of estimating gains from trade.

2.1 Relation Between DFS (1977) and EK (2002)

To gain insight into the EK model, let's establish a connection between EK and the DFS model. We'll demonstrate that the DFS model is a special case of the EK model, arising when there are two countries and relative efficiencies A adopt a particular functional form.

Formally, consider two countries (H) and (F), where a tilde will indicate foreign variables. Furthermore, we will also consider a continuum of goods, whose set is given by $J := [0, 1]$.

Recall that the DFS model describes the technology of production using unit labor requirements. In contrast, the EK model relies on the marginal productivity of labor, which conveys the same information when there is a single production factor. Specifically, the relation between both concepts is $z = \frac{1}{a}$, where a denotes unit labor requirements and z the marginal productivity labor.

EK respectively interprets each country's labor productivity, z and $\widetilde{z}$ as realizations of random variables $\underline{z}$ and $\underline{\widetilde{z}}$. This feature prevents us from making deterministic predictions about a specific draw. In other words, the model is silent about a country's efficiency for a particular good.

Despite this limitation, the model can still describe a country's overall production patterns. This is possible due to the existence of an infinite number of goods, by applying some Law of Large Numbers. Roughly, this law states that the average of a large number of draws will converge to its population mean. In the context of the EK model, the assumption of an infinite number of goods ensures enough draws so that the expected outcome actually represents the actual outcome observed. This allows the model to make predictions about a country's overall production.

As EK's ultimate goal is to use the model empirically, they assume specific distributions for the random marginal productivities. Specifically, they suppose that each $\underline{z}(j)$ and $\underline{\widetilde{z}}(j)$ is drawn from a Fréchet distribution, and so the cdfs are respectively

$$F(z) := \exp\left[-Tz^{-\theta}\right],$$
$$\widetilde{F}(\widetilde{z}) := \exp\left[-\widetilde{T}\widetilde{z}^{-\theta}\right],$$

where the parameters T and $\widetilde{T}$ are country specific, but θ is common across countries. In turn, the corresponding pdfs are

$$f(z) := \exp\left[-Tz^{-\theta}\right]\theta T z^{-\theta-1},$$
$$f(z) := \exp\left[-\widetilde{T}\widetilde{z}^{-\theta}\right]\theta \widetilde{T}\widetilde{z}^{-\theta-1}.$$

Focusing on country (H), a higher value of T indicates that (H) has higher efficiency draws on average. Thus, **the parameter T reflects absolute advantages**. Likewise, a higher θ means that draws are less dispersed, **turning θ a parameter capturing comparative advantages.** The intuition behind is that a high dispersion of productivity draws implies that the draws are more dissimilar on average. As a result, high dispersion is associated with larger differences in productivity across countries.

By restating the model in probabilistic terms, we can also reinterpret the function $A(j)$ from DFS. Given goods $J := [0,1]$, the value j can now be understood as the fraction of goods for which the expected relative efficiency is greater than some value A. Formally, $j = \Pr\left[\frac{\underline{z}}{\widetilde{z}} > A\right]$, and so

$$j = \frac{TA^{-\theta}}{TA^{-\theta} + \widetilde{T}}.$$

This makes it possible to derive $A(j)$, which is the relative efficiency of some given fraction of goods:

$$A(j) = \left(\frac{T}{\widetilde{T}}\right)^{\frac{1}{\theta}} \left(\frac{1-j}{j}\right)^{\frac{1}{\theta}}.$$

Suppose some value A. We want to know the fraction of goods such that the relative productivity of countries is A or more. Formally, this means, $j = \Pr\left[A < \frac{\widetilde{a}}{\underline{a}}\right]$. We now express it in terms of marginal productivities,

$$\begin{aligned} j &= \Pr\left[A < \frac{\widetilde{a}}{\underline{a}}\right], \\ &= \Pr\left[A\widetilde{z} < \underline{z}\right], \\ &= 1 - \Pr\left[\underline{z} < A\widetilde{z}\right], \\ &= 1 - \int_{\widetilde{z}} \exp\left[-T\widetilde{z}^{-\theta}A^{-\theta}\right] \mathrm{d}\widetilde{F}. \end{aligned}$$

Noting that $\widetilde{f}(\widetilde{z}) = \exp\left[-\widetilde{T}\widetilde{z}^{-\theta}\right]\theta\widetilde{T}\widetilde{z}^{-\theta-1}$,

$$\Rightarrow j = 1 - \int_{\widetilde{z}} \exp\left[-T\widetilde{z}^{-\theta}A^{-\theta}\right] \exp\left[-\widetilde{T}\widetilde{z}^{-\theta}\right]\theta\widetilde{T}\widetilde{z}^{-\theta-1}\mathrm{d}\widetilde{z}$$

$$\Rightarrow 1 - j = \widetilde{T}\int_{\widetilde{z}} \exp\left[-T\widetilde{z}^{-\theta}A^{-\theta} - \widetilde{T}\widetilde{z}^{-\theta}\right]\theta\widetilde{z}^{-\theta-1}\mathrm{d}\widetilde{z}$$

Noting that $\frac{TA^{-\theta}+\widetilde{T}}{TA^{-\theta}+\widetilde{T}} = 1$,

$$\begin{aligned} \Rightarrow 1 - j &= \widetilde{T}\int_{\widetilde{z}} \exp\left[-\left(TA^{-\theta}+\widetilde{T}\right)\widetilde{z}^{-\theta}\right]\theta\widetilde{z}^{-\theta-1}\frac{TA^{-\theta}+\widetilde{T}}{TA^{-\theta}+\widetilde{T}}\mathrm{d}\widetilde{z}, \\ &= \frac{\widetilde{T}}{TA^{-\theta}+\widetilde{T}}\int_{\widetilde{z}} \exp\left[-\left(TA^{-\theta}+\widetilde{T}\right)\widetilde{z}^{-\theta}\right]\theta\left(TA^{-\theta}+\widetilde{T}\right)\widetilde{z}^{-\theta-1}\mathrm{d}\widetilde{z} \end{aligned}$$

The integral is equal to 1, since the function to be integrated is the density of a Fréchet with constant $\left(TA^{-\theta}+\widetilde{T}\right)$. Thus, $1 - j^* = \frac{\widetilde{T}}{TA^{-\theta}+\widetilde{T}}$ and so

$$j^* = \frac{TA^{-\theta}}{TA^{-\theta}+\widetilde{T}}$$

Interpreting A as a function of j, we can invert the expression to obtain that:

$$1 - j = \frac{\widetilde{T}}{T\left[A(j)\right]^{-\theta}+\widetilde{T}}$$

$$\Rightarrow T\left[A(j)\right]^{-\theta} = \frac{\widetilde{T}}{1-j} - \widetilde{T}$$

$$\Rightarrow A(j) = \left(\frac{\widetilde{T}}{T}\frac{j}{1-j}\right)^{\frac{-1}{\theta}}$$

$$\Rightarrow A(j) = \left(\frac{\widetilde{T}}{T}\right)^{\frac{-1}{\theta}} \left(\frac{j}{1-j}\right)^{\frac{-1}{\theta}}$$

Given this expression for $A(j)$, the model becomes isomorphic to DFS. To have a truly equivalent model, we also need to add the trade-balanced condition, which remains unchanged. Then, country (H) will produce all goods in the interval $[0, j^*]$, where j^* satisfies $A(j^*) = \frac{w}{\widetilde{w}}$. The only difference relative to DFS is that j^* is now the fraction of goods that (H) is *expected* to produce.

Keep in mind that, when we work with a continuum of goods, the ***expected*** fraction of goods to be produced by (H) coincides with the fraction of goods that are ***actually produced***. This is because, even though there is uncertainty about whether (H) produces a good, given a continuum of goods there is no uncertainty in the aggregate. As we have mentioned, this is an implication of the Law of Large Numbers. For this reason, all the results in this model regarding aggregate variables can be taken as deterministic, rather than stochastic.

3 The EK Model

Building on the intuitions from the DFS, we next describe the EK model. We start by describing the setup, and then derive several probabilistic distributions describing the trade patterns.

3.1 Supply Side

There is a discrete set of countries $\mathcal{C} := \{1, 2, ..., C\}$, and a continuum of goods $\Omega := [0, 1]$. Each good $\omega \in \Omega$ is homogeneous and produced under perfect competition. Labor is the only production factor, which we suppose mobile within countries but not between them.[3]

In contrast to DFS, EK incorporates stochastic technology for producing goods. This technology exhibits constant returns to scale and is country-specific, with all firms having access to the same technology and receiving the same draw of labor unit requirement. Given the single production factor, unit labor requirements are inversely related

[3] Although labor is immobile between countries, EK somewhat relax the assumption of input mobility by considering the existence of intermediate inputs. Specifically, EK consider that each good produced can be either consumed or employ to produce another good. Since any produced good is tradable, the assumption implies that the intermediate inputs are effectively mobile between countries. Furthermore, our analysis focuses on a single production factor. However, the model can be generalized to accommodate multiple factors, a possibility that also exists in DFS. This requires assuming constant returns to scale technologies.

to marginal productivity. Consequently, we can fully characterize the technology by specifying a distribution for either concept. Following EK, but differing from DFS, we choose to specify the distribution through marginal productivity rather than unit labor requirements.

Formally, denote $\underline{z}_i := (\underline{z}_i(\omega))_{\omega\in[0,1]}$ the random vector describing country i's efficiency to produce goods. The random variable $\underline{z}_i(\omega)$ for good ω defines another random variable: the unit labor requirements $\underline{a}_i(\omega)$, such that $\underline{a}_i(\omega) = \frac{1}{\underline{z}_i(\omega)}$. The relation between both concepts arises since there is only one production factor. A specific draw of productivity for good ω is denoted $z_i(\omega)$. so that producing one unit of ω requires $a_i(\omega) = \frac{1}{z_i(\omega)}$ units of labor.

We suppose that each $\underline{z}_i(\omega)$ is drawn independently for each ω. The cdf of $\underline{z}_i(\omega)$ is denoted by F_i, which represents i's fraction of goods that have efficiency lower than a certain value. The distribution of $\underline{z}_i(\omega)$ is supposed to be Fréchet:

$$F_i(z_i(\omega)) := \exp\left(-T_i(z_i(\omega))^{-\theta}\right),$$

where $\theta > 1$. Notice that we are implicitly assuming that the distribution is not good-specific (neither T_i nor θ depend on ω). Consequently, the distribution of efficiency for each good is the same within a country.[4]

> There are several distributions widely used in the Trade literature. One of them is the Fréchet distribution. Its main property is that the maximum of n random variables distributed Fréchet is also Fréchet. This will be relevant for the determination of prices paid by country for a product, identified by the cheapest source country.

The random variable describing the unitary cost of a good ω in country i is $\underline{c}_i(\omega)$. Given draws of productivity in each country, the cost of producing one unit of ω in country i is:

$$c_i(\omega) = a_i(\omega) w_i = \frac{w_i}{z_i(\omega)}.$$

The random price of a firm from i in j is given by

$$\underline{p}_{ij}(\omega) = \frac{w_i}{\underline{z}_i(\omega)}\tau_{ij}. \tag{1}$$

Given realizations of efficiencies in each country, we can identify the price $p_{ij}(\omega)$ that

[4]Notice that the random variable representing efficiency is the same for all goods, but, ex post, different sectors are going to get different draws.

each country i sets in j. This in turn determines the price of the good ω in country j:

$$p_j(\omega) := \inf\{p_{ij}(\omega) : i \in \mathcal{C}\}.$$

Finally, delivering one unit of ω to country j when the good is produced in $i \neq j$ is subject to an iceberg trade cost τ_{ij}. Consequently, τ_{ij} units need to be sent from i to j in order to have one unit arriving in j. We adopt the convention that there are no costs in the domestic market, so that $\tau_{ii} = 1$, and that trade costs are symmetric, so that $\tau_{ij} = \tau_{ji}$ with $\tau_{ij} > 1$.

3.2 Demand Side

Preferences are identical in each country and given by a symmetric CES utility function:

$$U\left[(q_\omega)_{\omega\in[0,1]}\right] := \left[\int_0^1 (q_\omega)^{\frac{\sigma-1}{\sigma}} \mathrm{d}\omega\right]^{\frac{\sigma}{\sigma-1}},$$

where $\sigma > 1$ and is referred to as the elasticity of substitution.

Later in the course, we will elaborate on the properties of the CES utility. At this point, the demand side is unimportant. In particular, this is unnecessary for identifying the production and trade patterns, which are entirely determining by the supply side. For our specific purposes, there are only two features of the CES that you need to know: the utility is symmetric regarding goods and the indirect utility can be expressed as real income.

The symmetry of the CES utility function implies that one unit of any good provides the same level of utility, regardless of the good in question. Hence, if the prices of two goods are equal, the demand and expenditure of both goods will be equal too. As for the indirect utility function, it'll correspond to our measure of welfare. Just like the case of Cobb Douglas, this can be expressed as income.[5] Specifically, denoting the indirect utility for country i by V_i,

$$V_i(w_i, \mathbb{P}_i) := \frac{Y_i}{\mathbb{P}_i},$$

where Y_i is total income and $\mathbb{P}_i$ is a price index. Note that Y will equal wages in this

[5] This is not a coincidence. Both utility functions are homothetic, and any homothetic utility function has an indirect utility function that can be represented as real income. Furthermore, the Cobb Douglas utility is a particular case of the CES that arises when $\sigma \to 1$.

model, since there will be no profits in equilibrium. The price index is specifically given by

$$\mathbb{P}_i := \left[\int_0^1 [p(\omega)]^{1-\sigma} \, \mathrm{d}\omega\right]^{\frac{1}{1-\sigma}}.$$

Remember that $\mathbb{P}_i$ represents the price of a single representative bundle. This basket is referred to as a quantity index, and is such that one unit of this index gives one unit of utility.

3.3 Distributions

Before analyzing the model and its implications, let's obtain a few statistical distributions. Their derivations are necessary for some of the calculations performed subsequently.

First, let's determine the distribution of $\underline{p}_{ij}(\omega)$, defined by (1). The term $G_{ij}(p;\omega) := \Pr\left[\underline{p}_{ij}(\omega) \leq p\right]$ indicates the probability that the good ω produced in country i has a price lower than p when delivered to j. Given the assumptions of the model, this expression takes the following form:

$$G_{ij}(p;\omega) = 1 - \exp\left[-T_i p^{\theta} (w_i \tau_{ij})^{-\theta}\right].$$

We want to get the expression for $G_{ij}(p;\omega) := \Pr\left[\underline{p}_{ij}(\omega) \leq p\right]$. Since $\underline{p}_{ij}(\omega) = \frac{w_i}{\underline{z}_i(\omega)}\tau_{ij}$, then

$$G_{ij}(p;\omega) := \Pr\left[\underline{p}_{ij}(\omega) \leq p\right]$$

$$\Rightarrow G_{ij}(p;\omega) = \Pr\left[\frac{w_i}{\underline{z}_i(\omega)}\tau_{ij} \leq p\right]$$

$$\Rightarrow G_{ij}(p;\omega) = \Pr\left[\frac{w_i}{p}\tau_{ij} \leq \underline{z}_i(\omega)\right]$$

$$\Rightarrow \Pr\left[\underline{z}_i(\omega) \leq \frac{w_i}{p}\tau_{ij}\right] = 1 - G_{ij}(p;\omega)$$

This implies that $F_i\left(\frac{w_i}{p}\tau_{ij}\right) = 1 - G_{ij}(p;\omega)$, so that

$$G_{ij}(p;\omega) = 1 - F_i\left(\frac{w_i}{p}\tau_{ij}\right)$$

.Since $F_i(z_i(\omega)) := \exp\left(-T_i [z_i(\omega)]^{-\theta}\right)$, then $F_i\left(\frac{w_i}{p}\tau_{ij}\right) = \exp\left[-T_i p^{\theta} (w_i\tau_{ij})^{-\theta}\right]$ and so

$$G_{ij}(p;\omega) = 1 - \exp\left[-T_i p^{\theta} (w_i\tau_{ij})^{-\theta}\right]$$

Now, let's determine the distribution of the random variable representing the price of good ω in j. Since goods are homogeneous, the price that consumers in j will pay for good ω is given by the minimum price:

$$\underline{p}_j(\omega) := \inf_{c\in\mathcal{C}} \left\{\underline{p}_{cj}(\omega)\right\}.$$

Respectively denoting the cdf and pdf of this variable by $G_j(p;\omega)$ and $g_j(p;\omega)$,

$$G_j(p;\omega) = 1 - \exp\left[-\Phi_j p^\theta\right],$$
$$g_j(p;\omega) = \exp\left[-\Phi_j p^\theta\right]\Phi_j \theta p^{\theta-1},$$

where $\Phi_j := \sum_{c\in\mathcal{C}} T_c (w_c\tau_{cj})^{-\theta}$. Notice that the distribution G_j is the same for any good ω and is Fréchet.

By definition $G_j(p;\omega) = \Pr\left[\inf_{c\in\mathcal{C}} \underline{p}_{cj}(\omega) \leq p\right]$ and refers to the event comprising the states where, given draws of efficiency in each country, at least one country sets the price p and any other country sets a price equal or higher than p.

For the calculations, it is easy to note that this event is the complement of the event in which all countries are setting a price equal or higher than p. Thus, given the independence of distributions,

$$\begin{aligned}G_j(p;\omega) = 1 - \Pr\left[\underline{p}_j(\omega) \geq p\right] \Rightarrow G_j(p;\omega) &= \Pr\left[\bigcap_{c\in\mathcal{C}}\left\{\underline{p}_{cj}(\omega)\geq p\right\}\right],\\ &= 1 - \prod_{c\in\mathcal{C}}\left[1 - G_{cj}(p;\omega)\right],\\ &= 1 - \prod_{c\in\mathcal{C}}\exp\left[-T_c p^\theta (w_c\tau_{cj})^{-\theta}\right],\\ &= 1 - \exp\left[\sum_{c\in\mathcal{C}} -T_c p^\theta (w_c\tau_{cj})^{-\theta}\right],\end{aligned}$$

and defining $\Phi_j := \sum_{c\in\mathcal{C}} T_c (w_c\tau_{cj})^{-\theta}$ then

$$G_j(p;\omega) = 1 - \exp\left[\Phi_j p^\theta\right]$$

which defines the density

$$g_j(p;\omega) = \frac{dG_j(p;\omega)}{dp} = \exp\left[-\Phi_j p^\theta\right]\Phi_j\theta p^{\theta-1}.$$

Next, we calculate the probability that the random price of a good ω produced in i results in the minimum price in country j. Formally, this is $\pi_{ij} := \Pr\left[\underline{p}_{ij}(\omega) \leq \inf_{c\in\mathcal{C}\setminus\{i\}} \underline{p}_{cj}(\omega)\right]$, and provides information about the likelihood that country i serves country j. It is given by

$$\pi_{ij} = \frac{T_i (w_i\tau_{ij})^{-\theta}}{\Phi_j} = \frac{T_i (w_i\tau_{ij})^{-\theta}}{\sum_{c\in\mathcal{C}} T_c (w_c\tau_{cj})^{-\theta}}. \tag{2}$$

Calculating $\pi_{ij} := \Pr\left[\underline{p}_{ij}(\omega) \leq \inf_{c\in\mathcal{C}\setminus\{i\}} \underline{p}_{cj}(\omega)\right]$ requires considering all the possible realizations of draws that the country i can have, since $\underline{p}_{ij}(\omega)$ is a random variable. In addition, we also need to consider the draws that all the other countries could get, since each $\underline{p}_{cj}(\omega)$ is a random variable too.

To calculate π_{ij}, we proceed as follows. We start by considering a realization p of $\underline{p}_{ij}(\omega)$. The probability that p is the lowest price is the probability that the prices of all the other countries are greater than p. Formally, this is the probability that $\underline{p}_{cj} \geq p$ for each $c \neq i$. After this, we consider all the possible values p that country i could get as a draw. Formally,

$$\begin{aligned}
\pi_{ij} &:= \Pr\left[\underline{p}_{ij}(\omega) \leq \inf_{c\in\mathcal{C}\backslash\{i\}} \underline{p}_{cj}(\omega)\right], \\
&= \int_{p\in[0,\infty)} \Pr\left[p \leq \inf_{c\in\mathcal{C}\backslash\{i\}} \underline{p}_{cj}(\omega)\right] \mathrm{d}G_{ij}, \\
&= \int_0^\infty \Pr\left[\bigcap_{c\in\mathcal{C}\backslash\{i\}} \left\{p \leq \underline{p}_{cj}(\omega)\right\}\right] \mathrm{d}G_{ij}, \\
&= \int_0^\infty \prod_{c\neq i} \Pr\left[p \leq \underline{p}_{cj}(\omega)\right] \mathrm{d}G_{ij}, \\
&= \int_0^\infty \prod_{c\neq i} \left[1 - G_{cj}(p;\omega)\right] \mathrm{d}G_{ij}, \\
&= \int_0^\infty \prod_{c\neq i} \exp\left[-T_c p^\theta (w_c\tau_{cj})^{-\theta}\right] \mathrm{d}G_{ij}, \\
&= \int_0^\infty \exp\left[-p^\theta \sum_{c\neq i} T_c (w_c\tau_{cj})^{-\theta}\right] \mathrm{d}G_{ij}.
\end{aligned}$$

Knowing that the density function g_{ij} is given by $g_{ij}(\omega;p) := \frac{\mathrm{d}G_{ij}(p;\omega)}{\mathrm{d}p}$:

$$g_{ij}(\omega;p) := \exp\left[-T_i p^\theta (w_i\tau_{ij})^{-\theta}\right] T_i (w_i\tau_{ij})^{-\theta} \theta p^{\theta-1}$$

Then,

$$\begin{aligned}
\pi_{ij} &= \int_0^\infty \exp\left[-p^\theta \sum_{c\neq i} T_c (w_c\tau_{cj})^{-\theta}\right] \mathrm{d}G_{ij}, \\
&= \int_0^\infty \exp\left[-p^\theta \sum_{c\neq i} T_c (w_c\tau_{cj})^{-\theta}\right] \exp\left[-T_i p^\theta (w_i\tau_{ij})^{-\theta}\right] T_i (w_i\tau_{ij})^{-\theta} \theta p^{\theta-1} \,\mathrm{d}p, \\
&= \theta T_i (w_i\tau_{ij})^{-\theta} \int_0^\infty \exp\left[-p^\theta \sum_{c\in\mathcal{C}} T_c (w_c\tau_{cj})^{-\theta}\right] p^{\theta-1} \,\mathrm{d}p, \\
&= \theta T_i (w_i\tau_{ij})^{-\theta} \int_0^\infty \exp\left[-p^\theta \Phi_j\right] p^{\theta-1} \,\mathrm{d}p.
\end{aligned}$$

Using $g_j(p;\omega) = \exp\left[-\Phi_j p^\theta\right] \Phi_j \theta p^{\theta-1}$, so that multiplying and dividing by Φ_j:

$$\pi_{ij} = \frac{T_i (w_i\tau_{ij})^{-\theta}}{\Phi_j} \underbrace{\int_0^\infty \theta\Phi_j \exp\left[-p^\theta\Phi_j\right] p^{\theta-1} \,\mathrm{d}p}_{=1}$$

Finally, let's determine the distribution of the price paid by j for a good produced in i, when country i is the cheapest source in j. Formally, this is

$$H_{ij}(p) := \Pr\left[\underline{p}_{ij}(\omega) \leq p \,\middle|\, \underline{p}_{ij}(\omega) \leq \inf_{c\in\mathcal{C}\backslash\{i\}} \underline{p}_{cj}(\omega)\right],$$

and it can be shown that

$$H_{ij}(p) = G_j(p). \tag{3}$$

Let's show that $H_{ij}(p) = G_j(p)$. By definition

$$H_{ij}(p) := \Pr\left[\underline{p}_{ij}(\omega) \leq p \,\middle|\, \underline{p}_{ij}(\omega) \leq \inf_{c\in\mathcal{C}\setminus\{i\}} \underline{p}_{cj}(\omega)\right]$$

$$\Rightarrow H_{ij}(p) := \frac{\int_0^p \Pr\left[\bigcap_{c\in\mathcal{C}\setminus\{i\}} \left\{q \leq \underline{p}_{cj}\right\}\right] g_{ij}(q)\,\mathrm{d}q}{\pi_{ij}}$$

We know that

$$g_{ij}(\omega; p) := \exp\left[-T_i p^\theta (w_i\tau_{ij})^{-\theta}\right] T_i (w_i\tau_{ij})^{-\theta} \theta p^{\theta-1}$$

and also

$$\Pr\left[\bigcap_{c\in\mathcal{C}\setminus\{i\}} \left\{q \leq \underline{p}_{cj}\right\}\right] = \prod_{c\neq i} [1 - G_{cj}(q;\omega)]$$

which gives

$$\Pr\left[\bigcap_{c\in\mathcal{C}\setminus\{i\}} \left\{q \leq \underline{p}_{cj}\right\}\right] = \exp\left[-q^\theta \sum_{c\neq i} T_c (w_c\tau_{cj})^{-\theta}\right]$$

Therefore,

$$\begin{aligned} H_{ij}(p) &= \frac{1}{\pi_{ij}} \int_0^p \exp\left[-q^\theta \sum_{c\neq i} T_c (w_c\tau_{cj})^{-\theta}\right] \exp\left[-T_i q^\theta (w_i\tau_{ij})^{-\theta}\right] T_i (w_i\tau_{ij})^{-\theta} \theta q^{\theta-1}\,\mathrm{d}q, \\ &= \frac{1}{\pi_{ij}} \int_0^p \exp\left[-q^\theta \sum_{c\in\mathcal{C}} T_c (w_c\tau_{cj})^{-\theta}\right] T_i (w_i\tau_{ij})^{-\theta} \theta q^{\theta-1}\,\mathrm{d}q, \end{aligned}$$

and by multiplying and dividing by Φ_j:

$$H_{ij}(p) = \frac{1}{\pi_{ij}} \underbrace{\frac{T_i (w_i\tau_{ij})^{-\theta}}{\sum_{c\in\mathcal{C}} T_c (w_c\tau_{cj})^{-\theta}}}_{=\pi_{ij}} \underbrace{\int_0^p \exp\left[-q^\theta \sum_{c\in\mathcal{C}} T_c (w_c\tau_{cj})^{-\theta}\right] \sum_{c\in\mathcal{C}} T_c (w_c\tau_{cj})^{-\theta} \theta q^{\theta-1}\,\mathrm{d}q}_{=G_j(p)}.$$

Equation (3) has several implications for the conclusions of the model. First, each country i could have a different parameter T_i, and so different absolute advantages. However, conditional on serving the market j, any country would charge the same price in country j, on average. This follows because $H_{ij}(p)$ does not depend on i, since it equals $G_j(p)$ for any $i \in \mathcal{C}$. In other terms, the origin of the good has no bearing in expectation on the price determination.

Additionally, the distribution of prices charged in j by each country is the same. Due to the symmetry of utility, this implies that the distribution of quantities sold in j by each country i is also the same. This determines that, each country has the same distribution of prices and of quantities sold in j (both conditional on serving the market). A key implication of this is that, conditional on serving j, the expenditure on each good by any consumer from j is the same in expectations. As a result, the total expenditure share of country j in goods from i coincides with the fraction of goods produced in i and sold in j. Formally,

$$\frac{E_{ij}}{E_j} = \pi_{ij},$$

where E_{ij} is the expenditure of country j on goods produced by i and E_j is the total expenditure of country j.

4 Taking the EK Model to the Data

We have observed that one significant contribution of EK is its ability to connect the model with real-world data. When a model allows for this possibility, we usually classify it as **structural**. In essence, quantifying structural models requires estimating certain parameters, which can then be used to quantify the impact of economic shocks.

Specifically, EK can be used to get predictions about how changes in trade costs affect:

[1] Welfare

[2] Bilateral trade

To make the model empirically tractable, we need to reformulate certain components and express them in terms of observable variables. This involves deriving a set of equations that are functions of measurable quantities, allowing us to compute the model's predictions given a set of data.

Next, we show how certain equations can be rewritten for this purpose.

4.1 Welfare

We use the indirect utility function to measure welfare. To do this, we begin by getting an expression for the price index. For country i, and assuming $1+\theta-\sigma>0$, this is given by

$$\mathbb{P}_i = \left[\Gamma\left(\frac{1-\sigma+\theta}{\theta}\right)\right]^{\frac{1}{1-\sigma}} (\Phi_i)^{-\frac{1}{\theta}} ,$$

where Γ is the Gamma function.[6]

The price index of a CES is given by $(\mathbb{P}_i)^{1-\sigma} = \int_0^1 [p_i(\omega)]^{1-\sigma} \mathrm{d}\omega$. We use the distribution of prices to perform the integral. We know that the price distribution has cdf $G_i(p;\omega) = 1-\exp\left(-\Phi_i p^\theta\right)$ and pdf $g_i(p;\omega) = \exp\left(-\Phi_i p^\theta\right)\Phi_i\theta p^{\theta-1}$. So

$$(\mathbb{P}_i)^{1-\sigma} = \int_0^1 [p_i(\omega)]^{1-\sigma} \mathrm{d}\omega = \int_p t^{1-\sigma} g_i(t)\,\mathrm{d}t$$

[6] The Gamma function is given by $\Gamma(z) := \int_0^\infty x^{z-1}\exp(x)\,\mathrm{d}x$. It is a generalization of factorials to real numbers, so that if z is an integer, then $\Gamma(z) = (z-1)!$. More generally, if z is a real number, Γ satisfies that $\Gamma(z+1) = z\Gamma(z)$.

$$\Rightarrow (\mathbb{P}_i)^{1-\sigma} = \int_p t^{1-\sigma} \exp\left(-\Phi_i t^{\theta}\right) \Phi_i \theta t^{\theta-1}\, \mathrm{d}t$$

.

We express the results in terms of the Gamma function, defined by $\Gamma(z) := \int_0^{\infty} x^{z-1} \exp(x)\, \mathrm{d}x$. To do this, let $u := \Phi_i t^{\theta}$ so that $du = \Phi_i \theta t^{\theta-1} \mathrm{d}t$. Notice that $t = \left(\frac{u}{\Phi_i}\right)^{\frac{1}{\theta}}$. Then,

$$\begin{aligned}
(\mathbb{P}_i)^{1-\sigma} &= \int_p \left(\frac{u}{\Phi_i}\right)^{\frac{1-\sigma}{\theta}} \exp(-u)\, \mathrm{d}u, \\
&= \left(\frac{1}{\Phi_i}\right)^{\frac{1-\sigma}{\theta}} \int_p u^{\left(\frac{1-\sigma}{\theta}+1\right)-1} \exp(-u)\, \mathrm{d}u,
\end{aligned}$$

and assuming that $1+\theta-\sigma>0$, then

$$\begin{aligned}
\Rightarrow (\mathbb{P}_i)^{1-\sigma} &= \left(\frac{1}{\Phi_i}\right)^{\frac{1-\sigma}{\theta}} \Gamma\left(\frac{1+\theta-\sigma}{\theta}\right), \\
&= (\Phi_i)^{\frac{-1}{\theta}} \left[\Gamma\left(\frac{1+\theta-\sigma}{\theta}\right)\right]^{\frac{1}{1-\sigma}}
\end{aligned}$$

Ultimately, we do not care about the value of $\mathbb{P}_i$ itself, but rather its change when two different scenarios are considered. Broadly speaking, it is easier to express the change of a variable in percentage terms when we have multiplicative terms. Since the first term of the right-hand side of $\mathbb{P}_i$ plays no role (it acts as a constant), it is convenient to define $\gamma := \left[\Gamma\left(\frac{1-\sigma+\theta}{\theta}\right)\right]^{\frac{1}{1-\sigma}}$ and reexpress the price index as:

$$\mathbb{P}_i = \gamma (\Phi_i)^{-\frac{1}{\theta}}.$$

Likewise, the indirect utility function in country i is $\frac{w_i}{\mathbb{P}_i}$, which can be expressed by

$$\frac{w_i}{\mathbb{P}_i} = \gamma^{-1} (T_i)^{\frac{1}{\theta}} (\pi_{ii})^{\frac{-1}{\theta}}. \tag{4}$$

To get the expression, we start from $\pi_{ii} = \frac{T_i w_i^{-\theta}}{\Phi_i}$. Using that $\left(\frac{\mathbb{P}_i}{\gamma}\right)^{-\theta} = \Phi_i$, then $\pi_{ii} = \frac{T_i (w_i)^{-\theta}}{\left(\frac{\mathbb{P}_i}{\gamma}\right)^{-\theta}}$ which implies that $\pi_{ii} = \gamma^{-\theta} T_i \left(\frac{w_i}{\mathbb{P}_i}\right)^{-\theta}$ and the result follows.

In autarky, we know that $\pi_{ii}^{aut} := 1$. Therefore, we can use the indirect utility expressed in the form (4) to get the change in welfare relative to autarky:

$$\frac{w_i/\mathbb{P}_i}{(w_i/\mathbb{P}_i)^{aut}} = (\pi_{ii})^{-1/\theta}.$$

Notice π_{ii} can be easily obtained from the official statistics collected by countries. Thus, given an estimation of θ, we can estimate the welfare gains of the current situation relative to autarky.

4.2 Bilateral Trade: The Gravity Equation

The bilateral trade predicted by the model obeys the gravity equation:

$$E_{ij} = \frac{(\mathbb{P}_j)^{\theta}}{\sum_{c\in\mathcal{C}}\left(\frac{\mathbb{P}_c}{\tau_{ic}}\right)E_c}\left(\tau_{ij}\right)^{-\theta}E_jY_i. \tag{5}$$

We start from the equation $\frac{E_{ij}}{E_j} = \pi_{ij} = \frac{T_i(w_i\tau_{ij})^{-\theta}}{\Phi_j}$ so that $E_{ij} = \frac{T_i(w_i\tau_{ij})^{-\theta}}{\Phi_j}E_j$.
Using that $Y_i = \sum_{c\in\mathcal{C}} E_{ic}$, then

$$Y_i = \sum_{c\in\mathcal{C}} \frac{T_i\left(w_i\tau_{ic}\right)^{-\theta}}{\Phi_c}E_c$$

$$\Rightarrow \frac{Y_i}{\sum_{c\in\mathcal{C}}\frac{(\tau_{ic})^{-\theta}E_c}{\Phi_c}} = T_i\left(w_i\right)^{-\theta}$$

We plug in the expression for $T_i\left(w_i\right)^{-\theta}$ in the equation for E_{ij}, so that

$$E_{ij} = T_i\left(w_i\right)^{-\theta}\frac{\left(\tau_{ij}\right)^{-\theta}E_j}{\Phi_j}$$

$$\Rightarrow E_{ij} = \frac{Y_i}{\sum_{c\in\mathcal{C}}\frac{(\tau_{ic})^{-\theta}E_c}{\Phi_c}}\frac{\left(\tau_{ij}\right)^{-\theta}E_j}{\Phi_j}$$

Finally, the expression for the price index we found above determine that $\left(\frac{\mathbb{P}_c}{\gamma}\right)^{-\theta} = \Phi_c$, and so

$$E_{ij} = \frac{Y_i}{\sum_{c\in\mathcal{C}}\frac{(\tau_{ic})^{-\theta}E_c}{\left(\frac{\mathbb{P}_c}{\gamma}\right)^{-\theta}}}\frac{\left(\tau_{ij}\right)^{-\theta}E_j}{\left(\frac{\mathbb{P}_j}{\gamma}\right)^{-\theta}}$$

which reordering becomes

$$\Rightarrow E_{ij} = Y_iE_j\frac{\left(\frac{\mathbb{P}_j}{\tau_{ij}}\right)^{\theta}}{\sum_{c\in\mathcal{C}}\left(\frac{\mathbb{P}_c}{\tau_{ic}}\right)^{\theta}E_c}$$

5 Computation and Counterfactuals

For welfare calculations, it is not necessary to identify *all* equilibrium values: given an estimation of θ, the term π_{ii} can be directly obtained through data. Thus, all the terms in (4) can be easily computed.

Similarly, the gravity equation might be used without need to determine the equilibrium. For instance, EK uses the log version of (5) to provide an estimation of θ. By parametrizing trade costs and treating the rest of the variables as fixed effects, they do not need to solve for the equilibrium.

However, if our goal is to use the model to compare an observed situation relative to an unobserved counterfactual, we need to solve for the equilibrium of the model. This requires establishing the remaining equilibrium conditions. With this goal, we use that each country's income has to equal its expenditure:

$$w_i L_i = E_i.$$

The total income of country i is given by $w_i L_i$ and determined by the total revenue generated in the economy. Total revenue equals the value of exports and domestic sales, which are given by $\sum_{j\in\mathcal{C}} E_{ij}$. Since $E_{ij} = \pi_{ij} E_j$, we get that:

$$w_i L_i = \sum_{j\in\mathcal{C}} \pi_{ij} E_j.$$

Thus, wages can be solved by using the following system of equations:

$$w_i L_i = \frac{\sum_{j\in\mathcal{C}} T_i \left(w_i \tau_{ij}\right)^{-\theta} w_j L_j}{\sum_{c\in\mathcal{C}} T_c \left(w_c \tau_{cj}\right)^{-\theta}}. \tag{6}$$

Equation (6) follows by using that:
$w_i L_i = \sum_{j\in\mathcal{C}} \pi_{ij} E_j$
$\Rightarrow w_i L_i = \sum_{j\in\mathcal{C}} \pi_{ij} w_j L_j$
which relies on that $E_j = w_j L_j$. This determines $\pi_{ij} = \frac{T_i (w_i \tau_{ij})^{-\theta}}{\sum_{c\in\mathcal{C}} T_c (w_c \tau_{cj})^{-\theta}}$.

Once that wages for each country are obtained, variables such as trade flows and price indices can be obtained. Notice that using the model quantitatively supposes available estimations or calibrated values for $(T_i, L_i)_{i\in\mathcal{C}}$, θ, σ, and $(\tau_{ij})_{i,j\in\mathcal{C}}$.